

Prince Gabrielle Jhon M. Libertad

princegabriellejhonl@gmail.com • +63 960 346 9262 • +63 926 869 6011
[Portfolio](#) • [LinkedIn](#) • github.com/ASTRALLIBERTAD • Manila, Philippines

Professional Summary

Computer Engineering student focused on systems programming, compiler design, and hardware–software integration. Experienced in Rust and Python development across systems-level projects, including a custom compiler targeting LLVM IR and IoT-based environmental monitoring systems. Interested in offline AI deployment, embedded systems, and real-time data processing pipelines connecting low-level computation with user-facing applications.

EDUCATION

Polytechnic University of the Philippines (PUP)

Bachelor of Science in Computer Engineering | Manila, Philippines

2025 – Present

PROJECTS

MicroShell-ID — Marine Micro Gastropod Identification Tool *Faculty-supervised research project (PUP) | Ongoing International collaboration (Taiwan & Philippines)*

- Developed a prototype marine micro gastropod identification system integrating a Flutter desktop interface with AmScope MU1000 USB microscope hardware via UVC APIs.
- Engineered a high-performance Rust backend utilizing an ONNX inference model and a hybrid AI-morphological character decision engine.
- Implemented a robust Flutter–Rust FFI bridge to support secure, fully offline deployment in remote field environments.
- Participating in an ongoing faculty-supervised research collaboration; repository access is restricted pending institutional review.

Brain — Custom Compiled Programming Language (Compiler in Rust)

- Designed and implemented a custom programming language compiler in Rust with a multi-stage architecture including lexical analysis, parsing, and AST generation.
- Developed a backend code generation pipeline targeting LLVM Intermediate Representation (LLVM IR).
- Implemented core compiler infrastructure including tokenization, syntax parsing, and intermediate representation emission.
- Ongoing systems project focused on extending language features and optimization passes.

LMS Alternative — Learning Management System

- Python app with Google Drive content storage and Firebase authentication
- [Repo: github.com/ASTRALLIBERTAD/LMS-alternative](https://github.com/ASTRALLIBERTAD/LMS-alternative)

NH3 Monitoring System — BLE Ammonia Sensor

- ESP32 BLE sensor with native Kotlin Android app for real-time ammonia monitoring
- [Repo: github.com/ASTRALLIBERTAD/NH3-Monitoring-System](https://github.com/ASTRALLIBERTAD/NH3-Monitoring-System)

CERTIFICATIONS & TRAINING

Huawei ICT Academy

- Python Programming Basics — January 2026 (Code: EBG20260108002690)
- Algorithm and Program Design — September 2025 (Code: ICT20250919000003)

TECHNICAL SKILLS

Languages: Rust, Python, Kotlin, Dart (Flutter), C++

Frameworks & Libraries: Flutter, TensorFlow, ONNX Runtime, Firebase, Android SDK

Tools & Platforms: Git, CMake, Ninja, LLVM IR, Google Drive API, BLE / UVC protocols

Domains: Computer Vision, IoT & Embedded Systems, Compiler Design, Mobile Development